

1. Protocols Susceptible to Sniffing

- **Protocols that transmit data in plaintext** are highly susceptible to sniffing attacks because data can be captured and read easily.
 - Examples include:
 - **HTTP** (no encryption)
 - **FTP** (transmits credentials in clear text)
 - **Telnet** (no encryption)
 - **POP3 and IMAP** (email retrieval without encryption)
 - **DNS** (usually unencrypted)
 - **SNMP v1/v2** (no encryption)
 - Encrypted protocols like HTTPS, SSH, and FTPS protect against sniffing.
-

2. Active and Passive Sniffing

- **Passive Sniffing:**

-

The attacker silently listens to network traffic without altering it.

- Works well on networks using hubs or wireless where all traffic is broadcast to all nodes.

- **Active Sniffing:**

- The attacker actively interferes with the network to redirect traffic to themselves.
- Includes techniques like **ARP poisoning**, **MAC flooding**, or **man-in-the-middle attacks**.
- Common on switched networks where traffic is usually sent directly between devices.

3. ARP Poisoning

- **Address Resolution Protocol (ARP)** resolves IP addresses to MAC addresses.
 - **ARP poisoning (or spoofing)** involves sending fake ARP messages to a local network.
 - The attacker associates their MAC address with the IP address of another device (e.g., the default gateway), intercepting or altering traffic.
 - This is a common **man-in-the-middle (MITM)** attack that facilitates sniffing or session hijacking.
-

4. Ethereal Capture and Display Filters

- Ethereal (now Wireshark) is a popular packet capture and analysis tool.
 - Capture Filters: Applied before capturing traffic to limit which packets are saved.
 - Examples: host 192.168.1.1, port 80, tcp, udp.
 - Display Filters: Applied after capture to show only specific packets.
 - More powerful and flexible than capture filters.
 - Examples: ip.addr == 192.168.1.1, tcp.port == 80.
-

5. MAC Flooding

- A network attack targeting **network switches**.
- The attacker floods the switch with fake MAC addresses.
- This overwhelms the switch's MAC address table, causing it to fail open and broadcast all traffic to every port (like a hub).
- This allows the attacker to sniff traffic not normally visible.

6. DNS Spoofing Techniques

- Also called **DNS cache poisoning**.
- The attacker injects false DNS responses, redirecting users to malicious IP addresses.
- Can be performed by compromising a DNS server or intercepting DNS requests.
- Results in users visiting fake websites without their knowledge.

7. DNS Hacking

- Exploiting vulnerabilities in DNS servers or protocols.
- Common attacks include:
 - **DNS spoofing/cache poisoning**
 - **DNS amplification attacks** (used in DDoS)
 - **Zone transfer attacks** to obtain DNS zone data.
-

Aims to disrupt or manipulate domain name resolution.

8. Sniffing Countermeasures

- **Use Encrypted Protocols:** HTTPS, SSH, FTPS, and VPNs encrypt data in transit.
- **Static ARP Entries:** Prevent ARP poisoning by fixing IP-to-MAC mappings.
- **Port Security on Switches:** Limit MAC addresses per port to prevent MAC flooding.
- **Use Network Segmentation:** Isolate sensitive devices and traffic.
- **Use IDS/IPS:** Detect abnormal ARP traffic or MAC flooding attempts.
- **Employ DNSSEC:** To authenticate DNS responses and prevent spoofing.
- **Regularly update network devices:** To patch vulnerabilities.
- **Use strong authentication and authorization controls.**